

Problem set 7

Name: Diya Jain
U81673988

$$1. Q_1^+ = \bar{x} \text{ XOR } Q_1$$

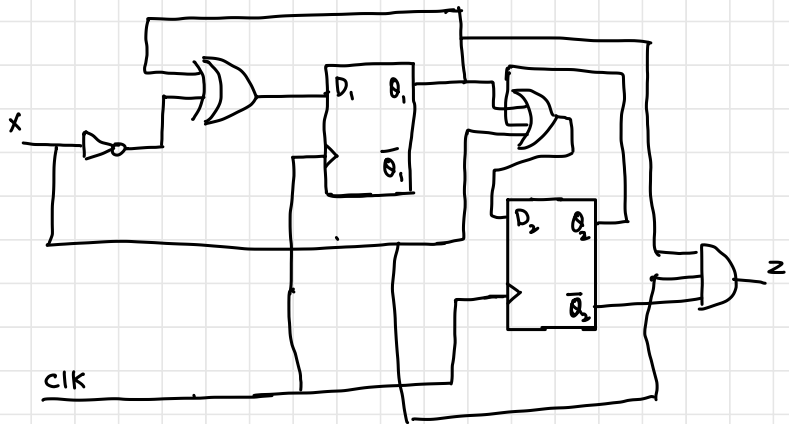
$$Q_2^+ = x + Q_1 + Q_2$$

$$z = x Q_1 Q_2'$$

Present input Next state flip flops Output

Q_1	Q_2	x	Q_1^+	Q_2^+	D_1	D_2	z
0	0	0	1	0	1	0	0
0	0	1	0	1	0	0	0
0	1	0	1	1	1	1	0
0	1	1	0	1	0	1	0
1	0	0	0	1	0	1	0
1	0	1	1	1	1	1	1
1	1	0	0	1	0	1	0
1	1	1	1	1	1	1	0

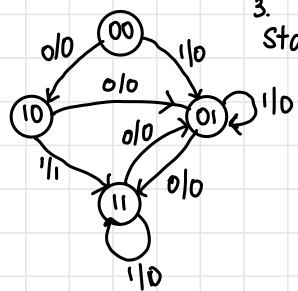
1. Logic diagram



2. Next state table

	x	
	0	1
$A = 00$	10	01
$B = 01$	11	01
$C = 11$	01	11
$D = 10$	01	11
	Q_1^+	Q_2^+

3. State transition diagram



2. a.

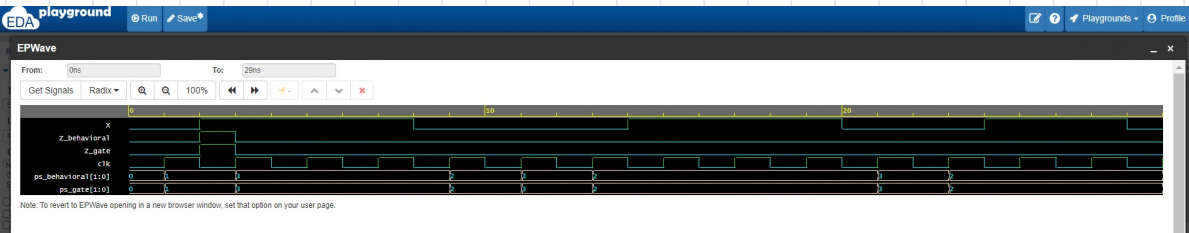
```
testbench.sv +
1 module Test_state_equations();
2   reg clk,X;
3   wire [1:0]ps_gate, ps_behavioral;
4   wire Z_gate,Z_behavioral;
5   Gate_level
6   G(clk,X,Z_gate,ps_gate);
7   Behavioral B(clk,X,Z_behavioral,ps_behavioral);
8
9   initial begin clk=0; X=0;
10  $dumpFile ("test.vcd");
11  $dumpvars;
12  #2 X=1;
13  #2 X=1;
14  #2 X=1;
15  #2 X=0;
16  #2 X=0;
17  #2 X=0;
18  #2 X=1;
19  #2 X=1;
20  #2 X=1;
21  #2 X=0;
22  #2 X=0;
23  #2 X=1;
24  #2 X=1;
25  #2 X=0;
26  #2 $finish;
27
28 end
29 always #1 clk = ~clk;
30
31 endmodule
```

```
design.sv +
1 //Part A
2 module Gate_level1(clk,X,Z,ps);
3   input clk,X;
4   output Z;
5   output [1:0]ps;
6   wire X_bar;
7   wire D1,D2,Q1,Q2,Q1bar,Q2bar;
8   not (X_bar,X);
9
10  xor (D1,X_bar,Q1);
11  or (D2,X,Q1,Q2);
12
13  buf (ps[1],Q2);
14  buf (ps[0],Q1);
15
16  D_flipflop ff1(clk,D1,Q1,Q1bar);
17  D_flipflop ff2(clk,D2,Q2,Q2bar);
18
19  and (Z,X,Q1,Q2bar);
20 endmodule
21
22 //D Flip Flop
23 module D_flipflop(clk,D,q,qbar);
24   input clk,D;
25   output q,qbar;
26   reg q=0;
27   assign qbar=~q;
28   always@(posedge clk) q<=D;
29 endmodule
30
```

b.

```
28 //Part B
29 module Behavioral1(clk,X,Z,ps);
30   input clk,X;
31   output reg Z;
32   output [1:0]ps;
33   reg [1:0]ps=0;
34   reg [1:0]ns;
35   parameter A=0, B=1, C=2, D=3;
36   always@(posedge clk)
37     ps<=ns;
38
39   always@(ps,X)
40     case(ps)
41       A:if(X)
42         begin ns=C; Z=0; end
43       else
44         begin ns=B; Z=0; end
45       B:if(X)
46         begin ns=D; Z=1; end
47       else
48         begin ns=C; Z=0; end
49       C:if(X)
50         begin ns=C; Z=0; end
51       else
52         begin ns=D; Z=0; end
53       D:if(X)
54         begin ns=D; Z=0; end
55       else
56         begin ns=C; Z=0; end
57       default: begin ns=A; Z=0; end
58     endcase
59 endmodule
60
```

C.

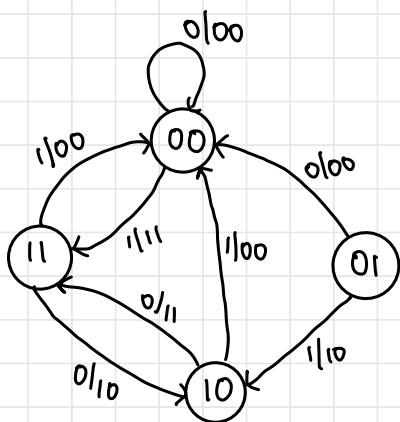
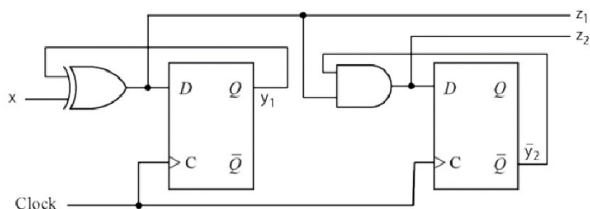


3.

input sequence = 0110111010

$$\begin{aligned} D_1 &= x \text{ XOR } y_1 \\ D_2 &= (x \text{ XOR } y_1) \bar{y}_2 \\ z_1 &= D_1 \\ z_2 &= D_2 \end{aligned}$$

y_1	y_2	x	D_1	D_2	y_1^+	y_2^+	z_1	z_2
0	0	0	0	0	0	0	0	0
0	0	1	1	1	1	1	1	1
0	1	0	0	0	0	0	0	0
0	1	1	1	0	1	0	1	0
1	0	0	1	1	1	1	1	1
1	0	1	0	0	0	0	0	0
1	1	0	1	0	1	0	1	0
1	1	1	0	0	0	0	0	0



input sequence: 0 1 1 0 1 1 1 0 1 0
 A C A A C A C D A A
 z_1, z_2 00 11 00 00 11 00 11 10 00 00
 z_1, z_2 0 1 0 0 1 0 1 0 0 0

4.

Run

Save*

Copy*

testbench.sv

SV/Verilog Testbench

```

1 module function_tb();
2
3   reg x, clk, nclr;
4   wire z1, z2;
5
6   Q4PS7 uu0(x, clk, nclr, z1, z2);
7
8   initial begin
9     clk = 1;
10    forever begin
11      #5 clk = ~clk;
12    end
13  end
14
15  initial begin
16    $dumpfile("test.vcd");
17    $dumpvars;
18  end
19
20  initial begin
21    nclr = 0;
22    #1 nclr = 1;
23  end
24
25  initial begin
26    $monitor("Time=%0t: x=%b, z1=%b, z2=%b", $time, x, z1, z2);
27    #5 x = 0;
28    #5 x = 1;
29    #5 x = 1;
30    #5 x = 0;
31    #5 x = 1;
32    #5 x = 1;
33    #5 x = 1;
34    #5 x = 0;
35    #5 x = 1;
36    #5 x = 0;
37    #55 $finish;
38  end
39
40 endmodule

```

design.sv

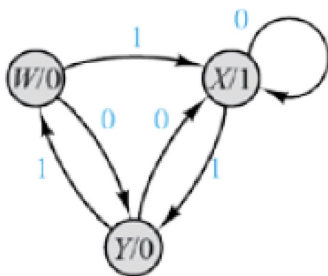
```

1 module DFF (d, clk, nclr, q);
2
3   input d, clk, nclr;
4   output reg q;
5
6   always @ (posedge clk or negedge nclr) begin
7     if (nclr == 1'b0) begin
8       q = 0;
9     end
10    else begin
11      q = d;
12    end
13  end
14
15 endmodule
16
17 module Q4PS7(x, clk, nclr, z1, z2);
18
19   input x, clk, nclr;
20   output z1, z2;
21
22   wire d1, d2;
23   wire y1, y2;
24
25   xor (d1, x, y1);
26   and (d2, d1, ~y2);
27
28   DFF u0(d1, clk, nclr, y1);
29   DFF u1(d2, clk, nclr, y2);
30
31   assign z1 = d1;
32   assign z2 = d2;
33
34 endmodule

```

Question 5 next page

5.



- a. from the above state transition diagram if there are sequence of zeros as input the output is one.
 and if the input is 10, output is 1.
 and if the input is 1100, output is 1.
 and if the input is 111, output is 1.

b.

	x=0	x=1
w	y/0	x/1
x	x/1	y/0
y	x/1	w/0

	x=0	x=1
w	11/0	01/1
x	01/1	11/0
y	01/1	00/0

x \ y ₂ y ₁	00	01	11	10
0	1	0	0	x
1	0	1	0	x

x \ y ₂ y ₁	00	01	11	10
0	1	1	1	x
1	1	1	0	x

$$D_1 = \bar{y}_2 \bar{x} + \bar{y}_1 y_2 x$$

$$D_2 = \bar{y}_1 + y_1 \bar{x}$$

